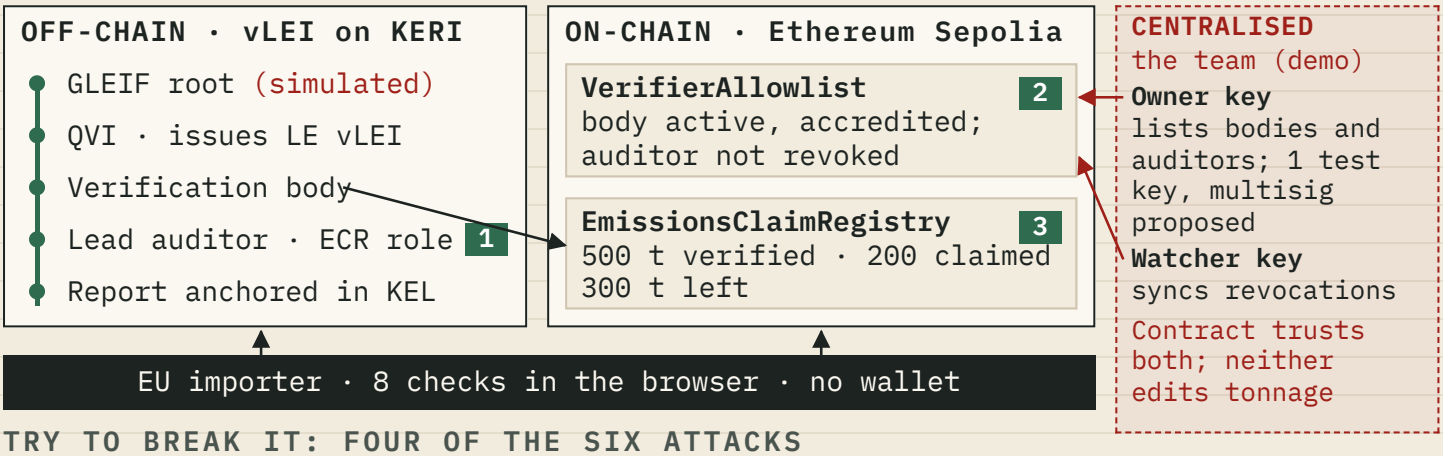
# A carbon number is only as trustworthy as the person who signed it.

**1 Who signed?**
The lead auditor's vLEI role credential (ECR), checked against a GLEIF-rooted chain (simulated root in the demo).

**2 Authorised at the registering block?**
At the block that registers the report, the contract checks that the body is active and accredited and the auditor is not revoked.

**3 Tonnes claimed once?**
**Within this deployment:** the ledger deducts each shipment from the 500 t verified and rejects a batch claimed twice or a claim beyond what remains.

## ARCHITECTURE




## TRY TO BREAK IT: FOUR OF THE SIX ATTACKS

1. **Tampered value** (1.8 edited to 1.2 tCO2e/t): **check 2 fails, DISCLOSURE_TAMPERED**; no contract is called.
2. **Tonnes claimed twice**, or 400 t when 300 t are left: **reverts BatchAlreadyClaimed / ExceedsVerifiedTonnage** (dry run).
3. **Auditor who left** (ECR revoked, watcher syncs): next report **reverts AuditorNotAuthorized on Sepolia**, status 0 · tx 0xa8b6b8a8…fc5
4. **Impostor body**, stolen owner key (simulated): **buyer's check 7 rejects it**; the contract had accepted it · tx 0x295abe6c…013

**REAL SCALE**

EU imports of CN 7318 from Taiwan, 2025 (Eurostat): **371,270 t**. Each 1% declared on a verified value instead of the 2026 default moves about **11,060 tCO2e** of declaration onto verifiers' signatures.

## MEASURED

**1,141** tests; fuzz, Slither, bounded Halmos + Maude model check in CI

**412/412** non-equivalent mutants killed

**19/19 · 0/28 · 4/4** tampered detected · valid rejected · borderline flagged, on inputs we wrote

**524,065** gas: one report + one claim (Sepolia gas has no value)

## WHAT IT IS NOT

- ✕ Not CBAM compliance
- ✕ Not a physical reduction
- ✕ Simulated GLEIF root
- ✕ Sepolia testnet only
- ✕ Fictional companies, people and LEIs

**TRY IT · SCAN FOR THE DEMO**

Demo, read-only, no wallet: **https://zuemen.github.io/carbon-lei/**
Repository: **https://github.com/zuemen/carbon-lei**
Real: Sepolia contracts, transactions, verifier, KERI signatures.